Quantitative Drift and Entropy Interfaces for Weak Hessian Bounds on RCD Spaces

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Abstract

The assigned manuscript passes from a distributional Hessian upper bound to a second-derivative inequality along every minimizing geodesic. We quantify two interfaces of its proof without replacing its nonsmooth geometric inputs. The original heat semigroup yields an explicit curvature-dependent beta-function bound for the drift resolvent. A spectral shift and the weighted Markov bound give a Lipschitz estimate at every positive resolvent parameter, together with coefficient and forcing error bounds in the correct measure. We also give a finite-parameter entropy-to-chord estimate that separately retains endpoint entropy, path concentration, varying curvature, and an allowed entropy defect. Quantitative graph closure and examples explain which finite errors can be propagated and which cannot. Nonbranching, Bochner measure calculus, and the curvature-dimension equivalences remain external inputs; no finite computation establishes the source's global theorem. Contribution assessment: C2/5, explicit source-specific analytic and entropy error bounds are proved, but the geometric assumptions and global theorem are not improved.

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1 The model and the actual proof route

Let (M, d, m) be a complete full-support $\mathrm{RCD}(K, N)$ space, $K \in \mathbb{R}$, $1 < N < \infty$, with finite measure on bounded sets. The source uses unreduced $\mathrm{CD}(K, N)$ plus quadratic Cheeger energy. The nonpositive generator satisfies

$$\int \Gamma(u, v)\,\mathrm{d}m = -\int (\Delta u) v\,\mathrm{d}m, \qquad \Gamma(u, v) = \langle \nabla u, \nabla v \rangle.$$

There is no introduced boundary condition or coordinate metric. The global test class consists of $g \in D(\Delta)$ with a bounded globally Lipschitz representative and $\Delta g \in W^{1,2}$. For locally Sobolev F, compactly supported such g, and $h \in \mathrm{Lip}_c(M)$, the source defines

$$H_F(\nabla g, \nabla g)(h) = -\int \mathrm{div}(h \nabla g) \Gamma(F, g)\,\mathrm{d}m - \int h\,\Gamma(F, \Gamma(g)/2)\,\mathrm{d}m. \tag{1.1}$$

This is a distributional evaluation, not an assumed pointwise Hessian. For bounded globally Lipschitz F and bounded continuous G, the assigned theorem asserts that

$$H_F(\nabla g, \nabla g)(h) \le \int h G \Gamma(g)\,\mathrm{d}m \quad (h \ge 0) \quad \Longrightarrow \quad (F \circ \sigma)'' \le \ell^2 G \circ \sigma \tag{1.2}$$

for every constant-speed minimizing $\sigma : [0, 1] \to M$ of length ℓ. No global $L^2(m)$ assumption on F is imposed. Our report analyzes the supplied route, not an independent proof of all of the external RCD theory on which it depends.

1.1 Source architecture and retained dependencies

All five scientific modules were freshly read in full.

1. `calculus.tex` constructs compact original generator cores, obtains graph-to-Hessian control, and admits Lipschitz multipliers against capacity-compatible Bochner measures.

2. `drift.tex` proves, for bounded Lipschitz V,

$$m_V = e^V m, \qquad L_V = \Delta_V = \Delta + \Gamma(V, \cdot), \qquad D(L_V) = D(\Delta). \tag{1.3}$$

   The graph norms are equivalent. Bounded multipliers with bounded weighted Laplacian are shown Lipschitz using the original heat flow.

3. `curvature-change.tex` proves on the compact original core the identity

$$\mathcal{B}_V(g, \xi) - \mathcal{B}_0(g, e^V \xi) = -H_V(\nabla g, \nabla g)(e^V \xi). \tag{1.4}$$

   It closes the integrated formula in the common generator graph domain. For $V = \lambda F$, constant and variable curvature-dimension equivalences then yield $\mathrm{CD}(K - \lambda G, \infty)$.

4. `every-geodesic.tex` uses properness and nonbranching to make a strict interior subsegment unique. It concentrates measures in endpoint balls and increases λ to eliminate entropy errors. Green-kernel inequalities on all subintervals give (1.2).

The Bochner multiplier calculus, Sobolev-to-Lipschitz principle, Bakry gradient estimate, nonbranching theorem, and the BE/CD equivalences are substantial imported results. Reading their use does not reproduce their full proofs. In particular, energy semigroup equivalence by itself supplies none of the every-geodesic passage.

2 Explicit drift-resolvent constants

All unmarked L^p norms in this section use m; essential suprema are unchanged by the bounded positive weight e^V. Put $D = \|\nabla V\|_\infty$ and $R_a = (a - \Delta)^{-1}$, $a > 0$. Define

$$I_K(t) = \int_0^t e^{2Ks}\,\mathrm{d}s, \qquad H_K(a) = \int_0^\infty \frac{e^{-at}}{\sqrt{2I_K(t)}}\,\mathrm{d}t. \tag{2.1}$$

The original Bakry estimate implies the classical reverse-Poincare inequality

$$2I_K(t)\Gamma(P_t h) \le P_t(h^2) - (P_t h)^2. \tag{2.2}$$

We retain this original-space analytic input, in the normalization of Savaré [2, Eq. (3.3)]; it is not a finite-matrix fact.

Lemma 2.1 (Two distinct resolvent bounds)**.** *For $h \in L^2 \cap L^\infty$,*

$$\|R_a h\|_p \le a^{-1}\|h\|_p \quad (p = 2, \infty), \qquad \|\nabla R_a h\|_2 \le \frac{\|h\|_2}{2\sqrt{a}}, \qquad \|\nabla R_a h\|_\infty \le H_K(a)\|h\|_\infty. \tag{2.3}$$

The explicit positive constants in (2.1) are

$$H_K(a) = \begin{cases} \dfrac{1}{2\sqrt{K}}\mathrm{B}\left(\dfrac{a+K}{2K}, \dfrac{1}{2}\right), & K > 0, \\ \sqrt{\dfrac{\pi}{2a}}, & K = 0, \\ \dfrac{1}{2\sqrt{-K}}\mathrm{B}\left(\dfrac{a}{-2K}, \dfrac{1}{2}\right), & K < 0. \end{cases} \tag{2.4}$$

They are finite for every $a > 0$ and tend to zero as $a \to \infty$.

Proof. The Markov contraction and the integral representation $R_a h = \int_0^\infty e^{-at} P_t h\,\mathrm{d}t$ give the first bounds. For the nonnegative spectral variable s of $-\Delta$, $\sqrt{s}/(a+s) \le 1/(2\sqrt{a})$, with equality at $s = a$. Spectral calculus proves the L^2 gradient estimate. This is the best bound from nonnegative spectrum alone, not a claim that every space has the maximizing spectral value.

For bounded data, (2.2) bounds $|\nabla P_t h|$ by $\|h\|_\infty/\sqrt{2I_K(t)}$. Integrate this bound; closedness of the weak differential justifies the integral, first over compact time intervals and then by convergence. With $K > 0$, substitute $u = e^{-2Kt}$ to obtain the first beta integral. For $K = -k < 0$, the same substitution $u = e^{-2kt}$ gives the third. For $K = 0$, use the gamma integral of $e^{-at}t^{-1/2}/\sqrt{2}$. Near zero the integrand is $O(t^{-1/2})$; at infinity it is bounded by a constant times e^{-at}, or decays faster if $K > 0$. Dominated convergence after fixing a positive lower value of a proves the last assertion. □

The source `drift.tex`, lines 59–88, uses an unspecified $C_K a^{-1/2}$ for large a. Formula (2.4) retains the curvature dependence of the same heat input. In particular the exact L^2 constant does not replace the independent L^∞ bound.

Proposition 2.2 (Forcing-series certificate)**.** *Set $T_a h = \Gamma(V, R_a h)$, $q_2 = D/(2\sqrt{a})$, and $q_\infty = D H_K(a)$. If both numbers are smaller than one, the solution of $(a - L_V)u = h$, $h \in L^2 \cap L^\infty$, is*

$$u = R_a \sum_{j=0}^{\infty} T_a^j h. \tag{2.5}$$

For truncation after $j = J$, its function error in each L^p, $p = 2, \infty$, is at most $a^{-1} q_p^{J+1}\|h\|_p/(1 - q_p)$. The corresponding gradient error replaces a^{-1} by $1/(2\sqrt{a})$ or $H_K(a)$.

Proof. The bounds on T_a follow directly from (2.3). The series converges in the complete intersection $L^2 \cap L^\infty$ with its sum norm. Its limit f obeys $f = h + T_a f$, so $u = R_a f \in D(\Delta) = D(L_V)$ solves the stated equation. The weighted energy identity proves uniqueness. Geometric tails and (2.3) prove the errors. This construction also supplies a Lipschitz representative by Sobolev-to-Lipschitz. □

No arbitrary constants are quotiented out in this proof. The positive parameter fixes the constant mode, and the forcing space rather than a possibly incomplete homogeneous gradient space carries the contraction.

3 All positive parameters and perturbations

The weighted form is Markovian, so its resolvent $R_a^V = (a - L_V)^{-1}$ obeys $\|R_a^V h\|_\infty \le \|h\|_\infty / a$. This estimate uses the actual measure m_V, not an unweighted symmetry assertion. Equivalent measures allow the same bounded data and the same essential supremum. Conservativity is not needed for this sub-Markov estimate.

Theorem 3.1 (Shifted multiplier bound)**.** *For every $a > 0$, define*

$$\mathcal{L}_{K,D}(a) = \inf_{\substack{b \ge a \\ DH_K(b) < 1}} \frac{bH_K(b)}{a[1 - DH_K(b)]}. \tag{3.1}$$

The infimum is finite. For $h \in L^2 \cap L^\infty$,

$$\operatorname{Lip}(R_a^V h) \le \mathcal{L}_{K,D}(a) \|h\|_\infty. \tag{3.2}$$

In particular, when $K \ge 0$, let $c = \sqrt{\pi/2}$. A bound is

$$\mathcal{L}_{K,D}(a) \le \begin{cases} 2\pi D/a, & 0 < a \le 2\pi D^2,\ D > 0, \\ c/(\sqrt{a} - cD), & a \ge 2\pi D^2, \end{cases} \tag{3.3}$$

and for $D = 0$ the bound is $c/\sqrt{a}$.

Proof. Put $u = R_a^V h$. It is bounded, in the common generator domain, and $L_V u = au - h$ is bounded. To avoid assuming Lipschitz regularity, choose a large b_0 for which both contraction bounds in Proposition 2.2 hold. Apply that proposition to $h + (b_0 - a)u \in L^2 \cap L^\infty$; uniqueness identifies its Lipschitz solution with u.

Now any $b \ge a$ with $DH_K(b) < 1$ is permitted, even if no separate L^2 contraction is asserted at that b. The original resolvent identity gives

$$u = R_b\{h + (b - a)u + \Gamma(V, u)\}.$$

All terms lie in $L^2 \cap L^\infty$. Taking gradient bounds and using the weighted Markov estimate yields

$$\operatorname{Lip} u \le H_K(b)\{(b/a)\|h\|_\infty + D \operatorname{Lip} u\}.$$

Absorb and take the infimum. For $K \ge 0$, $H_K(b) \le c/\sqrt{b}$. Write $t = \sqrt{b}$. The remaining minimization is that of $ct^2/[a(t - cD)]$ on $t \ge \sqrt{a}$, $t > cD$. Its derivative has the sign of $t - 2cD$, giving (3.3). At the joining point the formulas agree. This optimizes this particular bound, not the true operator norm on a general RCD space. □

Corollary 3.2 (Forcing and drift errors)**.** *Let V, W be bounded Lipschitz potentials on the same original space, $\delta = \|\nabla(V - W)\|_\infty$, and $u = R_a^V h$. Suppose $z \in D(\Delta)$ is bounded Lipschitz and $r = h - (a - L_W)z \in L^2 \cap L^\infty$. Then*

$$\begin{aligned} \|u - z\|_\infty &\le \frac{\|r\|_\infty + \delta \operatorname{Lip} z}{a}, \\ \operatorname{Lip}(u - z) &\le \mathcal{L}_{K,D}(a)(\|r\|_\infty + \delta \operatorname{Lip} z), \qquad D = \|\nabla V\|_\infty. \end{aligned} \tag{3.4}$$

Proof. The domain equality and the drift formula imply exactly $(a - L_V)(u - z) = r + \Gamma(V - W, z)$. Apply the weighted Markov estimate and Theorem 3.1. The source residual and the operator error are different terms, although the latter becomes a forcing after z is specified. □

These constants depend on the gradient of the potential, not on an arbitrary additive constant in it. A finite approximation still has to certify the full essential-supremum residual and drift error.

4 A quantitative version of graph closure

Fix a bounded Lipschitz V. Here every L^2 norm and integral uses the physical measure m_V. Let k be bounded and let $\xi \in D(L_V)$ be bounded Lipschitz with $L_V\xi$ bounded. For $g \in D(L_V)$, define the source's graph-form defect

$$\mathcal{Q}(g) = \frac{1}{2}\int L_V\xi\,\Gamma(g)\,\mathrm{d}m_V + \int L_V g\,\Gamma(\xi, g)\,\mathrm{d}m_V + \int \xi (L_V g)^2\,\mathrm{d}m_V - \int k\xi\Gamma(g)\,\mathrm{d}m_V. \tag{4.1}$$

The original proof needs nonnegativity for every admissible g, ξ, not just a finite list. The next statement estimates one comparison.

Proposition 4.1 (Graph-form error budget)**.** *For $g, z \in D(L_V)$, put $A_g = \|\nabla g\|_2$, $B_g = \|L_V g\|_2$, and likewise for z. Set $e = g - z$, $d_1 = \|\nabla e\|_2$, $d_2 = \|L_V e\|_2$. Then*

$$\begin{aligned} |\mathcal{Q}(g) - \mathcal{Q}(z)| \le{}& (\tfrac{1}{2}\|L_V\xi\|_\infty + \|k\xi\|_\infty) d_1(A_g + A_z) \\ &+ \operatorname{Lip}\xi\,(d_2 A_g + B_z d_1) + \|\xi\|_\infty d_2(B_g + B_z). \end{aligned} \tag{4.2}$$

If $e = R_a^V r$, $r \in L^2(m_V)$, one may use

$$\|e\|_2 \le \|r\|_2/a, \qquad d_1 \le \|r\|_2/(2\sqrt{a}), \qquad d_2 \le \|r\|_2. \tag{4.3}$$

Proof. The difference of the two squared gradient terms is bounded in L^1 by $d_1(A_g + A_z)$. In the mixed term write

$$L_V g\,\Gamma(\xi, g) - L_V z\,\Gamma(\xi, z) = L_V e\,\Gamma(\xi, g) + L_V z\,\Gamma(\xi, e).$$

Cauchy–Schwarz gives its displayed bound. The remaining square difference is at most $d_2(B_g + B_z)$ in L^1. For (4.3), use self-adjointness of $-L_V$ in $L^2(m_V)$: the three spectral multipliers are $(a + s)^{-1}$, $\sqrt{s}/(a + s)$, and $s/(a + s)$. Their suprema on $s \ge 0$ are $a^{-1}, (2\sqrt{a})^{-1}, 1$. □

This makes source `curvature-change.tex`, lines 117–134, quantitative for a specified residual. It does not turn finite tests into a proof of the BE inequality. If weighted errors are replaced by unweighted ones, measure factors must be retained. After harmlessly shifting V by a constant so $0 \le V \le \operatorname{osc} V$, $\|f\|_{2,m} \le \|f\|_{2,m_V} \le e^{\operatorname{osc} V/2}\|f\|_{2,m}$. For $V = \lambda F$, this may grow exponentially in λ. The qualitative graph-core argument is for each fixed λ; it asserts no uniform discretization of all those cores.

5 A finite entropy-to-chord inequality

Fix a constant-speed minimizing path $\tau : [0,1] \to M$ of length l. Let $0 < \varepsilon$, $B_i = B_\varepsilon(\tau_i)$, $\mu_i = m|_{B_i}/m(B_i)$, and let a probability π on constant-speed minimizing paths have these endpoint laws. Assume all its paths and τ lie in a compact Q, with $0 < m(Q) < \infty$, and satisfy

$$\sup_{s\in[0,1]} d(\gamma_s, \tau_s) \le \omega \quad \text{for } \pi\text{-almost every } \gamma. \tag{5.1}$$

The path lengths are at most $L_\varepsilon = l + 2\varepsilon$, and their differences from l are at most 2ε. Write

$$J_t = \frac{t(1-t)}{2}, \qquad g(s,t) = \min(s,t) - st, \qquad \omega_G(r) = \sup_{\substack{x,y\in Q \\ d(x,y)\le r}} |G(x) - G(y)|.$$

Thus $\int_0^1 g(s,t)\,\mathrm{d}s = J_t$, with maximum $1/8$, not $1/4$. Let $\mu_t = (e_t)_\# \pi$, and set

$$C_t = \int\!\!\int g(s,t)|\dot\gamma|^2\,\mathrm{d}s\,\mathrm{d}\pi, \qquad D_t = \int\!\!\int g(s,t)G(\gamma_s)|\dot\gamma|^2\,\mathrm{d}s\,\mathrm{d}\pi.$$

For one $\lambda > 0$, suppose the chosen plan satisfies the following possibly defective weighted entropy inequality, with finite entropies:

$$\mathrm{Ent}_{m_\lambda}(\mu_t) \le (1-t)\,\mathrm{Ent}_{m_\lambda}(\mu_0) + t\,\mathrm{Ent}_{m_\lambda}(\mu_1) - KC_t + \lambda D_t + \mathfrak{d}_t, \qquad \mathfrak{d}_t \ge 0. \tag{5.2}$$

Here $m_\lambda = e^{\lambda F} m$. For the source CD plan one may take $\mathfrak{d}_t = 0$. An arbitrary numerical path mixture does not automatically satisfy (5.2), even if its endpoints or pointwise distances have been matched.

Theorem 5.1 (Retained finite-parameter errors)**.** *Under these hypotheses, let $L_F = \operatorname{Lip} F$, $M_G = \sup_Q |G|$, $K_- = \max(0, -K)$, and*

$$\mathcal{H}_t = (1-t)\log\frac{m(Q)}{m(B_0)} + t\log\frac{m(Q)}{m(B_1)}. \tag{5.3}$$

Then

$$\begin{aligned} F(\tau_t) &\ge (1-t)F(\tau_0) + tF(\tau_1) - l^2\int_0^1 g(s,t)G(\tau_s)\,\mathrm{d}s - \mathcal{E}_t, \\ \mathcal{E}_t &= L_F(\varepsilon+\omega) + J_t\{L_\varepsilon^2\omega_G(\omega) + 4M_G\varepsilon(l+\varepsilon)\} \\ &\quad + \frac{\mathcal{H}_t + K_-L_\varepsilon^2 J_t + \mathfrak{d}_t}{\lambda}. \end{aligned} \tag{5.4}$$

The entropy budget is unchanged when m is multiplied by a positive constant. All errors refer to the same plan and same weighted law.

Proof. Put $A_t = \int F\,\mathrm{d}\mu_t$, $E_t = \mathrm{Ent}_m(\mu_t)$. Using $\mathrm{Ent}_{m_\lambda}(\mu) = \mathrm{Ent}_m(\mu) - \lambda\int F\,\mathrm{d}\mu$, (5.2) rearranges exactly as

$$A_t \ge (1-t)A_0 + tA_1 - D_t + \lambda^{-1}\{E_t - (1-t)E_0 - tE_1 + KC_t - \mathfrak{d}_t\}.$$

Jensen's inequality on Q gives $E_t \ge -\log m(Q)$, while $E_i = -\log m(B_i)$. Also $0 \le C_t \le L_\varepsilon^2 J_t$. These facts give the last line of (5.4). The tube bound yields $|F(\tau_t) - A_t| \le L_F\omega$; the endpoint error in the chord is at most $L_F\varepsilon$. Finally, pointwise along each path,

$$\begin{aligned} ||\dot\gamma|^2 G(\gamma_s) - l^2 G(\tau_s)| &\le L_\varepsilon^2\omega_G(\omega) + M_G||\dot\gamma|^2 - l^2| \\ &\le L_\varepsilon^2\omega_G(\omega) + 4M_G\varepsilon(l+\varepsilon). \end{aligned}$$

Integrating against the nonnegative Green kernel proves the result, including for signed G. The ratios in (5.3) prove normalization invariance directly. □

Corollary 5.2 (A usable parameter budget)**.** *Suppose* $m(B_\varepsilon(\tau_i)) \geq c_i\varepsilon^n$ *for* $0 < \varepsilon \leq \varepsilon_0$*, where* $n > 0$*,* $c_i > 0$*, and suppose an independently justified tube modulus* $\omega(\varepsilon)$ *is known. Let* $\mathfrak{d}_t \leq A_0 + \lambda\eta$ *uniformly in* t*. For any* $\zeta > 0$*, taking*

$$\lambda \geq \frac{n\log(1/\varepsilon) + \log(m(Q)/\min(c_0, c_1)) + K_-L_\varepsilon^2/8 + A_0}{\zeta} \tag{5.5}$$

when the numerator is positive makes the last line of (5.4) *at most* $\zeta + \eta$*.*

Proof. Bound the convex combination of endpoint log-ratios by their larger upper bound and use $J_t \leq 1/8$. If the displayed numerator is nonpositive, no additional positive lower bound on λ is needed. □

Local Bishop–Gromov comparison can supply ball-volume lower bounds after their constants and enclosing region are fixed; no universal value of c_i is asserted. More importantly, nonbranching alone is not an explicit numerical tube modulus. The source obtains convergence of all shrinking-endpoint paths by properness and uniqueness of each strict interior subsegment. Its compactness argument supplies no rate.

5.1 The ordered limit and its geometric gate

For a strict subsegment τ, suppose $\varepsilon_j \downarrow 0$, $\omega_j \to 0$, and

$$\frac{1 + \max_{t\in[0,1]} \mathcal{H}_t^j}{\lambda_j} \to 0, \qquad \sup_t \frac{\mathfrak{d}_t^j}{\lambda_j} \to 0.$$

Uniform continuity of G on the fixed compact region makes every term in (5.4) vanish. This recovers the source chord inequality. Repeating it on every strict subinterval, not only one interval or one finite grid, implies the distributional inequality: subtract a function whose second derivative is $\ell^2 G \circ \sigma$ and use the characterization of concavity by all chords.

For clarity, uniqueness of a strict interior subsegment follows from metric nonbranching as follows. Replacing it by a different minimizer would give two minimizing paths between the original endpoints, of the same length, sharing a nontrivial initial interval. Nonbranching rules this out. Properness gives compactness of uniformly bounded constant-speed paths. Every shrinking-endpoint limit is therefore the chosen subsegment; otherwise a subsequence bounded away from it would give a different minimizer. These are metric arguments, not properties of the drift-resolvent contraction. Deng's theorem providing nonbranching in this RCD setting is retained as an external input.

6 Examples and negative controls

Example 6.1 (A certified large-parameter choice). For $K \geq 0$, $D = 1$, and $a = 1$, direct absorption with $H_0(1) = \sqrt{\pi/2}$ fails. The shifted bound is nevertheless $\operatorname{Lip} R_1^V h \leq 2\pi\|h\|_\infty$. With a fully rational upper constant $c \leq 4/3$, choose $b = 64/9$, which gives $\operatorname{Lip} R_1^V h \leq 64\|h\|_\infty/9$. This is an upper estimate, not a sharp multiplier norm or a new regularity theorem.

Example 6.2 (Actual circle drift). On the circle of circumference 2π with arclength metric and measure, take $V(x) = v\cos x$, $z(x) = \cos x$. Then

$$L_V z = -\cos x + v\sin^2 x, \qquad (a - L_V)z = (a+1)\cos x - v\sin^2 x.$$

Replacing v by w changes the forcing by $-(v-w)\sin^2 x$; it is not the additive constant $V - W$ that enters the differential operator. Self-adjointness is in $L^2(e^{v\cos x}\,\mathrm{d}x)$. Ignoring that weight in a graph norm would certify a different operator. These identities hold for every real v, w and $a > 0$, including $v = 0$.

Example 6.3 (A finite chord budget). Take a Euclidean unit segment, $\varepsilon = 1/100$, $\omega \le \varepsilon$, $K = 0$, $L_F \le 1$, $M_G \le 1$, and $\omega_G(r) \le r$. Linear interpolation of any endpoints in the two balls verifies this tube bound in Euclidean space. At $t = 1/2$, the nonentropy error is at most

$$\frac{1}{50} + \frac{1}{8}\left\{\left(\frac{51}{50}\right)^2 \frac{1}{100} + 4\frac{1}{100}\frac{101}{100}\right\} = \frac{52701}{2000000}.$$

If a genuine entropy plan also has $\mathcal{H}_t \le 20$, $\mathfrak{d}_t \le 1$, and $\lambda = 1000$, its total error is at most $94701/2000000 < 1/20$. The stipulated entropy and curvature hypotheses are not inferred from the endpoint geometry.

Example 6.4 (Why endpoints alone do not select a path). Antipodal points on a circle have two minimizing semicircles. Concentrating endpoints at those two points does not force path laws onto a specified semicircle: paths can converge to the other one, whose midpoint has positive distance from the chosen midpoint. This is not a counterexample to the source. Such a whole antipodal semicircle is not a strict interior subsegment of a longer minimizing geodesic. The interior restriction and nonbranching step are essential.

Example 6.5 (Why one chord or one jet is insufficient). The polynomial $f(t) = t^2(t-1/2)^2(t-1)^2$ agrees with the zero function at $0, 1/2, 1$, but $f''(1/2) = 1/8 > 0$. Passing zero-curvature chord tests at those three points does not establish concavity. More generally a smooth bump supported away from a finite set has every jet zero at that set but may have positive second derivative elsewhere. This is an obstruction to finite sampling, not to a certificate with a proved continuum remainder.

The source already warns that the test class need not be preserved: bounded Lipschitz V agreeing with $-|x|$ and smooth g agreeing with x give $L_V g = -\operatorname{sign} x$ locally, not a $W^{1,2}$ function. We retain that example and do not claim its rediscovery as a contribution. Likewise the exact spectral constant $1/(2\sqrt{a})$ is classical and is not, by itself, a Lipschitz or every-geodesic theorem.

7 Original Ram comparison and limitations

Fresh original TR8 reading [7], lines 81–240, covers the coercive weak inverse, the operator realization, Green equivalence with a boundary lifting, resolvent perturbation identities, and spectral heat/Duhamel equivalence. The complete heat proof is in lines 196–236; the start of the wave section at 237–240 is not used as a completed wave proof. In that setting the pivot space is $L^2(\Omega)$, the energy space is $H_0^1(\Omega)$, and the inverse acts on the actual dual load. Its kernel is not automatically continuous, and its domain need not be $H^2 \cap H_0^1$. These cautions directly support retaining the domains in (1.3) and the physical measure in (4.1).

The weak-operator identity behind Corollary 3.2 is the same classical inverse-difference identity proved there, now applied to the RCD drift with the explicit multiplier bound already established. We do not substitute the Euclidean Dirichlet heat kernel for the RCD semigroup. In particular, TR8's energy equivalence does not supply (2.2), Sobolev-to-Lipschitz, a capacity-compatible Bochner measure, or nonbranching.

Original TR208 [8], fresh lines 143–251, specifies the strong regularity needed for differentiated jets and proves the multi-index Leibniz identity. An order-P output jet of an order-N operator generally needs input derivatives through $N + P$; the system remains rectangular until valid additional conditions close it. Here the drift is merely bounded and measurable as a vector field, arising from a Lipschitz potential. Differentiating it into a smooth coefficient matrix would discard the very domain issue the source addresses. We use the original jet proof to specify this obstruction, not to assert that RCD spaces admit smooth charts or that a finite jet determines a global multiplier.

7.1 What is improved, and what is not

The explicit beta constants, the optimized spectral shift, and the drift-error estimate expose parameters hidden by the source's "sufficiently large" resolvent argument. The graph-form estimate allows actual residual errors to be propagated without differentiating the drift. The finite entropy inequality makes the separate costs of concentration, entropy, curvature variation and a transport defect visible. These are useful conditional refinements with complete local proofs; none changes the source's curvature or regularity assumptions.

The analytic ingredients remain classical. We therefore assess this report at C2, independently of its L3 applicability, rather than claiming a new geometric theorem or a historically new resolvent method. The source already proves qualitative generator regularity, graph closure and the entropy limit. We quantify those interfaces; we do not count them as newly discovered arguments.

The costs are concrete. The shifted estimate grows like a^{-1} at small a when $D > 0$, and large λ increases the drift of $V = \lambda F$. Conversion of graph errors to an unweighted norm can additionally cost $e^{\lambda \operatorname{osc} F/2}$. A finite quadrature or sample has no automatic bound on the full residual. The entropy certificate needs a genuine admissible plan with the stated inequality and a certified tube modulus; its existence and modulus cannot be read from the finite formulas.

7.2 Extensions requiring additional input

One can improve the resolvent constant if a sharper original heat gradient estimate is available, for example in a specific smooth geometry. The shift proof then applies to its Laplace transform in place of H_K. No uniform optimality of (2.4) is claimed. An effective quantitative uniqueness theorem for the chosen geodesic would make $\omega(\varepsilon)$ in Theorem 5.1 explicit; properness and nonbranching alone provide only its limit. Unbounded potentials would require new integrability, domain and entropy arguments: bounded equivalence is used essentially here. Infinite-dimensional RCD spaces also require a replacement for the particular properness and nonbranching inputs. These are possible research directions, not proved extensions of the source theorem.

8 Public status and verification scope

Public primary checks were made on October 9, 2026. Brena–Gigli's arXiv v1 abstract and Remarks 2.3–2.4 were inspected [3]; they identify the weak-Hessian converse and the weighted regularization obstacle. Their full proof was not audited. Savaré's v2 statement of the gradient and reverse-Poincare estimates, Equations (3.2)–(3.3), was checked [2]; the underlying RCD self-improvement theorem remains imported. Deng's v2 abstract and journal metadata [4] state finite- dimensional metric nonbranching, and Braun–Habermann–Sturm's v1 abstract records the variable BE/CD equivalence [5]. These metadata and statement checks are not full-proof verification.

The public repository's family-356 scope note [6] describes a formalization of the every-geodesic theorem. We read that scope declaration, not the Lean proof or its axioms, and did not build it. It is not independent validation in this report. A fresh public path-specific GitHub history query returned the initial commit `adc7f124` dated October 6, 2026. Separately, the cached texts match the project's captured October 8 identity record. Neither statement converts byte provenance into scientific validity.

The owned ledger records all nine assigned public text files, including the five scientific modules, bibliography, wrapper, preamble and README: 1,037 literal lines with no unread complement in that inventory. Fresh private reads are TR8 81–240 and TR208 143–251, totaling 269 lines; the two inventoried private files have 1,816 unread lines, listed as exact

complementary intervals. Other private corpus files and the full external literature are outside the read claim. PRE hashes were saved before scientific displays, and POST hashes check stability. Discovery searches and truncated search output are excluded; no private text was transmitted online.

The owned script `M685-checks.py` passed 412 exact cases and 968 counted assertions. It checks polynomial evaluations of special beta integrals, spectral and shift formulas, actual circle drift identities, Neumann-domain polynomial graph defects on $[0, 1]$, Dirichlet Green integrals, signed curvature-transfer errors, geometric tails and the printed examples. The zero-grid polynomial and the domain-jump calculation are negative controls. There are no floating-point cases and no parent or foreign cases in these totals. Script SHA, runtime and groups are recorded. Exact finite algebra does not establish the RCD calculus, nonbranching, or a global curvature-dimension inequality.

9 Conclusion

The source's two difficult interfaces must remain distinct: weighted generator regularity is an analytic statement in a fixed measure, whereas selecting every prescribed geodesic is a geometric limiting argument. Explicit resolvent, graph and entropy error bounds make that distinction operational. They provide scoped quantitative information while leaving the source's global inputs visible.

References

[1] OpenAI, *Weak Hessian bounds along every geodesic in RCD spaces*, September 24, 2026. Pinned public manuscript. All assigned scientific text read; external global inputs retained.

[2] G. Savaré, *Self-improvement of the Bakry–Émery condition and Wasserstein contraction of the heat flow in RCD spaces*, Discrete Contin. Dyn. Syst. 34 (2014), 1641–1661. arXiv:1304.0643v2, May 17, 2013. Equations (3.2)–(3.3) inspected; full proof imported.

[3] C. Brena and N. Gigli, *Fine representation of Hessian of convex functions and Ricci tensor on RCD spaces*, Potential Anal. 62 (2025), 703–737. arXiv:2310.07536v1, October 11, 2023. Abstract and Remarks 2.3–2.4 inspected.

[4] Q. Deng, *Hölder continuity of tangent cones in RCD(K,N) spaces and applications to non-branching*, Geom. Topol. 29 (2025), 1037–1114. arXiv:2009.07956v2, October 4, 2020. Abstract/version record checked; full proof imported.

[5] M. Braun, K. Habermann and K.-T. Sturm, *Optimal transport, gradient estimates, and pathwise Brownian coupling on spaces with variable Ricci bounds*, J. Math. Pures Appl. 147 (2021), 60–97. arXiv:1906.09186v1, June 21, 2019. Abstract/version record checked; full proof imported.

[6] OpenAI Math repository, Family-356 formalization scope declaration. Accessed October 9, 2026; declaration only, no Lean rebuild or proof audit.

[7] M. Subbarao, confidential original TR8. Local directory `RamNonlin-IE-PDE-astra/TR8-lin-non-lin`; file `TR8-lin-non-lin.tex`, freshly read lines 81–240. Weak domains, Green/resolvent and energy heat proofs.

[8] M. Subbarao, confidential original TR208, local directory `res-rt/TechReps/FinalTechReps/TRs`; subdirectory `TR208-multidimensional-variable-coefficient-operators`. Same-stem TeX, freshly read lines 143–251; strong-jet regularity and exact differentiated operator identity, not a nonsmooth RCD chart theorem.